

September Crossword Solutions

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1 1	2 2
3 1	9

ONE DOWN

The total number of unrestricted paths from $(0, 0)$ to $(7, 7)$ is

$$\binom{14}{7} = 3432,$$

since each path consists of 7 steps right and 7 steps up.

Next, we subtract paths that enter the Pumpkin Patch. A path can only enter the patch from the left or bottom edges, so we count paths passing through each relevant edge/corner.

- Bottom-left corner $(2, 2)$: Paths reaching $(2, 2)$ take 2 right and 2 up steps, $\binom{4}{2}$ ways, and from there to $(7, 7)$ requires 5 right and 5 up steps, $\binom{10}{5}$ ways, giving a total of $\binom{4}{2} \times \binom{10}{5} = 1512$ paths.
- Other bottom-edge points (excluding bottom-left corner):
 - Enter at $(3, 2)$: To reach $(3, 2)$, take 3 right and 2 up steps, $\binom{5}{2}$ ways, subtract $\binom{4}{2}$ for paths that pass through $(2, 2)$, then multiply by $\binom{9}{5}$ to reach $(7, 7)$.
 - Enter at $(4, 2)$: To reach $(4, 2)$, take 4 right and 2 up steps, $\binom{6}{2}$ ways, subtract $\binom{5}{2}$ for paths that enter $(3, 2)$, then multiply by $\binom{8}{5}$ to reach $(7, 7)$.
 - Enter at $(5, 2)$: To reach $(5, 2)$, take 5 right and 2 up steps, $\binom{7}{2}$ ways, subtract $\binom{6}{2}$ for paths that enter $(4, 2)$, then multiply by $\binom{7}{5}$ to reach $(7, 7)$.

The sum for these three squares is

$$\left[\left(\binom{5}{2} - \binom{4}{2} \right) \binom{9}{5} + \left(\binom{6}{2} - \binom{5}{2} \right) \binom{8}{5} + \left(\binom{7}{2} - \binom{6}{2} \right) \binom{7}{5} \right] = 910.$$

Multiply by 2 to account for the symmetric left edge, giving $910 \times 2 = 1820$.

Adding the bottom-left corner contribution: $1820 + 1512 = 3332$ paths that enter the Pumpkin Patch.

Finally, the number of paths avoiding the Pumpkin Patch is

$$3432 - 3332 = 100 = 2^2 \times 5^2.$$

Hence, our answer is $(2 + 2) + (2 + 5) = \boxed{11}$.

THREE ACROSS

Denote the n -th numerator as

$$A_n = 10 + 10^3 + 10^5 + \cdots + 10^{2n-1}.$$

This is the geometric series

$$A_n = 10 \sum_{j=0}^{n-1} 10^{2j} = 10 \times \frac{10^{2n} - 1}{10^2 - 1} = \frac{10(10^{2n} - 1)}{99}.$$

We can denote the final series as

$$S = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{A_n}{101^n} = \frac{10}{99} \sum_{n=1}^{\infty} (-1)^{n+1} \frac{10^{2n} - 1}{101^n}.$$

Splitting the sum,

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{10^{2n}}{101^n} = \sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{100}{101} \right)^n.$$

This is a geometric series with ratio $-\frac{100}{101}$:

$$\sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{100}{101} \right)^n = \frac{\frac{100}{101}}{1 + \frac{100}{101}} = \frac{100}{201}.$$

Similarly,

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{101^n} = \frac{\frac{1}{101}}{1 + \frac{1}{101}} = \frac{1}{102}.$$

$$S = \frac{10}{99} \left(\frac{100}{201} - \frac{1}{102} \right).$$

Subtracting the inner fractions,

$$\frac{100}{201} - \frac{1}{102} = \frac{100 \times 102 - 201}{201 \times 102} = \frac{9999}{20502}.$$

Thus,

$$S = \frac{10}{99} \times \frac{9999}{20502} = \frac{1010}{20502} = \frac{505}{10251}.$$

$$\frac{m}{n} = \frac{505}{10251} \Rightarrow m + n = 505 + 10251 = 10756.$$

So, our answer is $1 + 0 + 7 + 5 + 6 = \boxed{19}$.

ONE ACROSS

Fix boy B_1 in one seat to remove rotational symmetry. Then consider the 6 remaining positions for boy B_2 (the second boy). The number of valid positions for B_3 , so that no two boys sit next to each other, is

$$4, 3, 3, 3, 3, 4$$

depending on where B_2 is seated around the circle. Summing these gives 20 possible placements for B_3 relative to B_2 . The other 6 people (the remaining adults and girls) can be arranged freely in

$$6! = 720 \text{ ways.}$$

Thus, the total number of seating arrangements is

$$24 \times 6! = 24 \times 720 = 14,400.$$

Since the total equals $100m$, we have

$$14400 = 100m \Rightarrow m = 144 \Rightarrow \sqrt{m} = \boxed{12}.$$

TWO DOWN

Notice that a 30-day month has five Saturdays if and only if its first day is Friday or Saturday. Let w_Y be the weekday of January 1 of year Y (mod 7, with Monday = 0 up to Sunday = 6). For year Y ,

$$\text{wd}(\text{Sep } 1) = \begin{cases} w_Y + 5 \pmod{7}, & \text{if } Y \text{ is common;} \\ w_Y + 6 \pmod{7}, & \text{if } Y \text{ is leap,} \end{cases}$$

with wd representing the numerical weekday (as defined above) of some date in a given year.

Thus, September has five Saturdays if and only if common $Y : w_Y \in \{6, 0\}$ or leap $Y : w_Y \in \{5, 6\}$. From one year to the next, $w_{Y+1} \equiv w_Y + 1$ for common and $w_{Y+1} \equiv w_Y + 2$ for leap. Between 2001 and 2099, there are no century-leap exceptions, so this process repeats every 28 years. In each 28-year block, exactly 8 years satisfy the condition.

Decomposing 2000-2099, we see that 2001-2028 (8 years), 2029-2056 (8 years), 2057-2084 (8 years), and 2085-2099 (5 years) all work. Year 2000 itself also qualifies as a leap year with Jan 1 being Saturday.

Hence, the total is $8 + 8 + 8 + 5 = \boxed{29}$ years.